

Embedded imperatives and command performatives with an embedded clause in Korean^{*}

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Jae-II Yeom. 2018. Embedded imperatives and command performatives with an embedded clause in Korean. *Language and Information* 22.2, 27-54. A command can be issued with an imperative or an explicit performative with an embedded clause. However, an explicit performative with an embedded imperative is not acceptable. If an imperative is embedded, it becomes a *ko*-phrase, where *ko* is a complementizer. If an explicit performative is used as a command, it takes a *kes*-phrase with a modal adnominal ending *-(u)l*. I claim that a *ko*-phrase is interpreted as an intensional semantic entity, which makes it possible for a *ko*-phrase to express a non-realistic content of a command. With a *kes*-phrase, an explicit performative can express a command because the *kes*-phrase can denote an extensional semantic entity as a realistic content of a command. In showing these differences, I propose the semantics of an imperative and an explicit performative as world changers. (Hongik University)

Key words: imperative, explicit performative, *ko*, *kes*, world changer

1. Introduction

In Korean, a full sentence ends with a mood marker. One property of Korean is that even when an utterance of a full sentence is reported, or conveyed, by a different speaker, a mood marker is retained in the form of a *ko*-phrase as a complement of a conveying verb.¹ Here are

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¹ In this paper, I use the following abbreviations: acc: accusative case marker, adn: adnominal ending, adni: imperfective adnominal ending, adnm: adnominal ending with the meaning of modality, adnp: perfective adnominal ending, cmp: complementizer, dec: declarative, exh: exhortative, impd: direct imperative, impi: indirect imperative, int: interrogative, nml: nominalizer, nom: nominative case marker, npst: non-past, prm: promissive, pst: past, top: topic marker, voc: vocative.

some examples:

- (1) inho-nun mina-ka ttena-ss-{ta, nya}-ko malha-yess-ta.
 Inho-top Mina-nom leave-pst-{dec, int}-cmp say-pst-dec
 'Inho said that Mina left.'
- (2) inho-nun mina-eykey tangcang ttena-{la, ca, ma}-ko malha-yess-ta.
 Inho-top Mina-to at.once leave-{imp, exh, prm}-cmp say-pst-dec
 'Inho {ordered, suggested, promised} that Mina {should, should, would} leave at once.'

In these examples, *-ta*, *-nya*, *-la*, *-ca*, and *-ma* are mood markers that represent sentence types. *-ko* is assumed to be a complementizer and it takes a Mood Phrase (= MP), which is headed by a mood marker. In these examples, the conveying verb in the root clause is *malha* 'say'.

There are various conveying verbs that reflect the sentence type of an embedded clause:

- (3) inho-nun mina-ka ttena-ss-{ta, nya}-ko {cwucangha, mwul}-(y)ess-ta.
 Inho-top Mina-nom leave-pst-{dec int}-cmp {assert, ask}-pst-dec
 'Inho {asserted that, asked if} Mina left.'
- (4) inho-nun mina-eykey tangcang ttena-{la, ca, ma}-ko {myenglyengha, ceyanha, yaksokha}-yess-ta.
 Inho-top Mina-to at.once leave-{imp, exh, prm}-cmp {order, propose, promise}-pst-dec
 'Inho {ordered, suggested, promised} Mina that she {should, should, would} leave at once.'

In these cases, uses of such conveying verbs are redundant because the sentence types of the embedded clauses are obvious.²

With an NP complement, however, the semantics of the conveying verb is important in interpreting the NP complement:

- (5) na-nun mina-ka ttena-ss-{um, nun-ci}-(l)ul {cwucangha, mwut}-(nu)n-ta.
 I-top Mina-nom leave-pst-{nml, adn-nml}-acc {assert, ask}-npst-dec
 'I {assert that, ask if} Mina left.'
- (6) na-nun ne-eykey tangcang ttena-l kes-ul {myenglyengha, ceyanha, l-top you-to at.once leave-adnm thing-acc {order, propose,

² *-ta* can be exceptional because a sentence ending with *-ta* can be embedded by various verbs that are not conveying verbs.

yaksokha}-n-ta.

promise}-npst-dec

'I {order, suggest, promise} you that you {should, should, would} leave at once.'

The sentences in (6) have the same meanings as the corresponding imperative, exhortative, and promissive that have the same contents as the embedded clauses:³

(7) tangcang ttena-{la, ca, ma}!

at.once leave-{imp, exh, prm}

'{Leave, Let's leave, I promise I will leave} at once!'

Interestingly enough, conveying verbs are not used as a performative verb if they take a *ko*-phrase. If the embedding verb is forced to be used as a performative verb, the sentence becomes odd:

(8) ??na-nun ney-ka ttena-ss-{ta, nya}-ko {cwucangha, mwut}-(nu)n-ta.

I-top you-nom leave-pst-{dec int}-cmp {assert, ask}-pst-dec

'I {assert that, ask if} you left.'

(9) ??na-nun ne-eykey tangcang ttena-{la, ca, ma}-ko {myenglyengha,

I-top you-to at.once leave-{imp, exh, prm}-cmp {order,

ceyanha, yaksokha}-n-ta.

propose, promise}-npst-dec

'I {order, suggest, promise} you that you {should, should, would} leave at once.'

These sentences are odd because they cannot be used as a performative or descriptive one. The sentences cannot be used descriptively because it is not natural for a speaker herself to

3 In this use, Korean has a special grand style ending of a declarative sentence:

i. na-nun ne-eykey tangcang ttena-l kes-ul myenglyengha-nola.

I-top you-to at.once leave-adnm thing-acc order-npst-dec

'I order you to leave at once.'

One property of that ending is that the subject is always the speaker:

ii. {nay, ??ney, ??mina}-ka o-ass-nola.

{I, ??You, ??Mina}-nom come-pst-dec

'{I, ??You, Mina} came.'

For this reason, it can be easily used in a performative sentence.

describe a situation of uttering a sentence with her own utterance. If the subjects of such sentences do not refer to the speakers, the sentences are fine if they are assumed to describe the situations in which the subjects of them convey a command, suggestion, or promise:

- (10) inho-nun mina-eykey tangcang ttena-{la, ca, ma}-ko {myenglyengha,
Inho-top Mina-to at.once leave-{imp, exh, prm}-cmp {order,
ceyanha, yaksokha}-n-ta.
propose, promise}-npst-dec
'Inho {orders, is ordering} Mina to leave at once.'

As far as I know, there is no previous literature mentioning these observations. I will explain why a sentence with a *ko*-phrase is not used as a performative sentence, while a sentence with a *kes*-phrase is.

This is a general phenomenon, but due to the limitation of the space, I will limit my discussions to imperatives and performatives that are used as commands (= command performatives). What is special about them is that the morphological form of an embedded imperative is slightly different from that of the root imperative. I suppose that this difference provides a clue to the explanation of the reason that a sentence with a *ko*-phrase is not used as a performative.

Before I discuss imperatives as root clauses or embedded clauses and command performatives, I will define terms. According to Austin (1962), a sentence is a performative if the speaker does another act in addition to the act of uttering the sentence. However, the term of performative can be ambiguous. It is said that an imperative is a performative, because it does not just describe what the actual state of affairs is like. An imperative cannot be true or false. When an imperative is uttered, in addition to the act of utterance, the speaker does something to give a command or a permission to the addressee. On the other hand, when we say that a verb is a performative verb, the use of the verb entails that there is also a speech act that is expressed by the verb itself. To distinguish the two meanings, I will call the latter an **explicit performative** and the former just a **performative** in its narrower sense. When I say command performative, it refers to an explicit performative used as a command.

The paper is organized as follows. In Section 2, I discuss conditions for two imperative mood markers in Korean and semantic and pragmatic meanings of imperatives. In doing this, I also discuss the meaning of the complementizer *ko*. In Section 3, I discuss pragmatic and semantic meanings of explicit performatives and the interpretation of a *kes*-phrase, and show how an explicit performative with a *kes*-phrase is understood as a command performative. In Section 4, I conclude the paper.

2. Imperatives and two imperative markers

2.1. *-(e)a)la* in root sentences vs. *-(u)la* in both root and embedded clauses

In Korean, it is well known that there are two imperative mood markers: *-(e)a)la* and *-(u)la*. See Nam and Ko (1985). An imperative is normally uttered with the addressee as the intended recipient, who is in the utterance context. The former form is a normal ending for imperatives as root clauses, but if an imperative is embedded, the latter form is used. I will call them **impd** and **impi**.

- (11) inho-ya, pap-ul chenchenhi mek-{ela, *ula}!
Inho-voc meal-acc slowly eat-{impd, impi}
'Inho, eat the meal slowly!'
- (12) mina-nun inho-eykey pap-ul chenchenhi mek-{*ela, ula}-ko malha-yess-ta.
Mina-top Inho-to meal-acc slowly eat-{impd, impi}-cmp say-pst-dec
'Mina told Inho to eat the meal slowly.'

The actual forms of the two morphemes vary with the preceding sound or morpheme. In certain phonological and morphological contexts, their forms can become the same:⁴

- (13) Impd: *-y(e)la*: after verb ha 'do'
 -la: after /e/ or /a/ that is not in ha 'do'⁵
 -(e, a)la: after a consonant
 -({e, a})la: other vowels than /e/ or /a/⁶
- Impi: *-la*: after a vowel or /l/
 -ula: after a consonant (including /l/)

Impi can have two forms when preceded by /l/, and both impd and impi have the same form when preceded by /e/ or /a/. One exception is that impd becomes *-y(e)la* after the verb

4 There are other forms with *-ke-la* or *-ne-la*. They are irregular. A verb like *talla* 'ask for' is itself imperative but it is considered a base-form. I will not discuss irregular cases here.

5 Here I do not use IPA phonetic symbols but scripting symbols. This is better because readers only see scripting symbols and they are also used to represent the sounds of words.

6 When impd begins with a vowel, the vowel becomes /e/ or /a/ depending on the vowel of the preceding syllable: if the vowel of the preceding syllable is /a/ or /o/, the form is *-ala*, and otherwise, the form is *-ela*.

~*ha* 'do'. Here I suppose that /y/ is inserted to prevent two vowels from being adjacent to each other.

One interesting fact is that *impi* can be used in root clauses too. (14) below can be uttered when the speaker thinks the intended addressee may not be in the utterance context. (14a) can be uttered as a demand in a political demonstration, without knowing whether the President is listening. In (14b), the speaker is asking Inho by radio to respond to the call, wondering whether Inho is listening. In uttering these sentences, the referents of the subjects are the intended addressees, but the speakers assume that they may not be listening to them. For this reason, it is odd to use a second person pronoun, as in (15).

- (14) a. *taythonglyeng-un tayhwa-ey ungha-{la, ??yla}*.
 President-top conversation-to consent-{*impi*, *impd*}
 '(I/We demand) President consent to a conversation!'
- b. *inho-nun untapha-{la, ??y(e)la}*. *ope*.
 Inho-top respond-{*impi*, *impd*} over
 'Inho respond to the call! Over.'
- (15) *??ne-nun taytapha-la!*
 you-top answer-*impi*
 'You answer me!'

This can be contrasted with (16). In (16a), the speaker knows (or, assumes) that Mina is listening, and in (16b) the subject is a second person pronoun.

- (16) a. *mina-nun taytapha-yla!*
 Mina-top answer-*impd*
 'Mina answer me!'
- b. *ne-nun taytapha-yla!*
 you-top answer-*impd*
 'You answer me!'

From these observations, we can get the following generalizations:

- (17) a. An imperative with *impd* is used if c_s assumes that c_A is listening.
 b. *Impi* is used in a root clause if the speaker assumes that the intended addressee may not be in the utterance context.
- (18) *Impd* can be used with a second person pronoun, but *impi* cannot.

The condition for *impd* is that it be used if the speaker assumes that the intended addressee is in the utterance context. On the other hand, *impi* is used if the speaker may not be in the utterance context. For this reason, *impi* cannot be used with a second person pronoun. This does not mean that an imperative with *impi* is uttered to no one. The speaker intends to convey the meaning of the sentence to the intended addressee if the addressee happens to hear the message. If not, the speaker wants his desire to talk to the intended addressee to be known to the addressee.⁷

There is another difference between imperatives with *impi* and imperatives with *impd*. There are two forms of vocatives:

- (19) a. direct vocative: *-(y)a* (/y/ is inserted after a vowel.)
 b. indirect vocative: *-(i)ye* (/i/ is inserted after a vowel.)⁸

The vocative forms depend on the styles of imperatives. The direct vocative form is used in plain style when the addressee is assumed to be in the utterance context, while the indirect vocative form is used in grand style when the addressee is not assumed to be in the utterance context:

- (20) a. *inho-{ya, ??ye}, ppalli o-ala!*
 Inho-{voc, voc} quickly come-impd
 ‘Inho, come quickly!’
 b. *inho-{ya, ??ye}, huymang-ul kaci-ela!*
 Inho-{voc, voc} hope-acc have-impd
 ‘Inho, have a hope!’
 (21) a. *chinkwu-{??ya, ye}, koi camtul-la! (on an epitaph)*
 friend-{voc, voc} peacefully sleep-impi
 ‘Friend, sleep peacefully!’
 b. *chengnyen-{??a, iye}, kkwum-ul kaci-la!*
 youth-{voc, voc} dream-acc have-impi
 ‘Boys, have a dream!’

7 Nam & Ko (1985) pointed out that *impi* can be used with a honorific marker *si*, while *impd* cannot. I suppose that this is not a significant difference. The reason is that the *impd* *-(e)la* is used only when the addressee is inferior to the speaker in their status or ages and a separate form is used when the addressee is superior to the speaker: *-seyyo* (= *-si-e-yo*) or *-si-psio*. On the other hand, there is no other variant for *impi* depending on the status or ages of the speaker and addressee.

8 In this morpheme, *-i* can be considered a copula for two reasons. One is that the honorific marker *si*, which is normally attached to a verb, can be inserted after *-i* and the preceding noun is often a common noun which heads a definite NP.

In an imperative with *impd*, the vocative is *-(y)a*, as shown in (20), while in an imperative with *impi*, the vocative is *-(i)ye*, as in (21). Even if a vocative is used, the conditions in (17) are valid. In (20), the speaker is directly talking to the friend. In (21a), the addressee does not exist in the actual world. In (21b), the addressee is a kind of people, which is defined across possible worlds and may not have a realization of that kind in the actual world. This shows that the two vocatives are distinctively used, depending on the conditions in (17) and (18).

The question is why *impi* can be used in an imperative as a root clause and in an embedded imperative. You might claim that imperatives with *impi* as root clauses correspond to embedded imperative clauses with the rest deleted. This is possible because embedded clauses are not based on local contexts:

- (22) *inho-nun na-eykey ne-lul manna-la-ko ha-yess-ta.*
Inho-top I-to you-acc meet-impi-cmp say-pst-dec
 'Inho told me to meet you.'

Assuming that a context is defined as a tuple of the speaker, the addressee, the utterance time and the common ground $\langle c_S, c_A, c_T, c_{CG} \rangle$, the local context for the embedded imperative in (22) is $\langle c_S = \text{Inho}, c_A = I, c_T, c_{CG} \rangle$. If the embedded imperative were interpreted with respect to this local context, *ne* 'you' would be understood as referring to the addressee (= "I") in the local context, which corresponds to the speaker in the utterance context for the root clause. However, it does not refer to the speaker, but the addressee, of the whole sentence.⁹

Since an embedded clause is also interpreted with respect to the utterance context for the whole sentence(, except the tense), we might assume that *impi* is used in an embedded clause because the addressee is assumed not to be in the utterance context for the root clause. This is exactly what we observed in (9), where we saw that the context for the embedded imperative is the same as that for the whole statement, and that the sentence becomes odd.

9 There is another piece of evidence that an imperative with *impi* as a root clause is not derived from an embedded imperative by deleting the rest of the original sentence. Imperatives with *impi* can be used with a vocative in root clauses, as shown in (21), while embedded imperatives cannot. (21) cannot be rephrased as follows:

- i. *??na-nun chinkwu-ye, koi camtul-la-ko malha-n-ta.*
I-top friend-voc peacefully sleep-impi-cmp say-npst-dec
 'I say that Friend, sleep peacefully.'

This indicates that *impi* can be used only if the speech act is an indirect one. Therefore, an imperative with *impi* as a root clause cannot be considered an embedded imperative with the rest deleted.

Contexts for embedded imperatives can be different from the utterance contexts for the root clauses in two ways:

- (23) a. The speaker of the embedded imperative is not a 1st person pronoun;
b. The utterance times of the embedded imperative and the whole sentence are different.

For example, if the speaker is not a 1st person, the sentence becomes fine, even if the addressee for the embedded clause is a second person:

- (24) inho-nun ne-eykey tangcang ttena-la-ko myenglyengha-n-ta.
Inho-top you-to at.once leave-imp-cmp order-npst-dec
'Inho orders you to leave at once.'

This sentence can be uttered in a context where the speaker reports a situation in which Inho is issuing a command to the addressee. If the tense is not the present, the sentence becomes fine, as in (25), even when the speaker is a 1st person and the addressee is a second person:

- (25) na-nun ne-eykey tangcang ttena-la-ko myenglyengha-yess-ta.
I-top you-to at.once leave-imp-cmp order-pst-dec
'I ordered you to leave at once.'

In (24) and (25), the contexts for the embedded clauses are different from the utterance contexts for the root clauses one way or another and we saw that sentences with embedded imperatives are not interpreted as command performatives. This indicates that a *ko*-phrase is not used when the context for the embedded clause is the same as that for the root clause.

The observations so far can be summarized as follows:

(26)

Root clauses		embedded clauses	
addressee present	addressee not present	local context = utterance context	local context $\neq$ utterance context
impd, *impi	*impd, impi	*impd, *impi	*impd, impi

A context where the addressee is not in the utterance context is a case where the local context is not the same as the utterance context, and *impi* is used in such a case. Then when the local context is the same as the utterance context, it is expected that *impd* is used, but it is not. Neither is *impi*. Therefore with an imperative mood marker in an embedded clause, the sentence is not used as a command performative.

The question is why no command performative is allowed if an imperative is embedded, as we saw in (9). However, this may not be the right question because neither *impd* nor *impi* is used in an embedded clause even if the context for the embedded clause is the same as the utterance context for the root clause. If an imperative is embedded, it becomes a *ko*-phrase. Therefore, the question should be why the embedding sentence cannot be a command performative with a *ko*-phrase. I suppose that it results from the basic meaning of *ko*. For this reason, I will look into the semantics of a *ko*-phrase.

2.2. Meanings of imperatives and *ko*-phrases

A sentence has a semantic meaning and various pragmatic meanings. Sometimes the two types of meanings interact with each other. If we do not know the semantic meaning of a sentence, we can determine it from its interactions with pragmatic meanings. If an utterance is made, the speaker is bound to follow the cooperative principle in the sense of Grice (1975).¹⁰ Any additional meanings we get from the assumption of the principle are called implicatures. Before I discuss imperatives, I will consider declaratives. Under the context $c = \langle c_S, c_A, c_T, c_{CG} \rangle$, if a declarative sentence is uttered, it is implicated that the speaker believes, and wants the addressee to believe, that the sentence is true, in addition to the fact that the speaker has uttered the sentence. But we do not expect the addressee to believe it:

(27) Mina is happy.

- | | |
|---|------------------------|
| (28) a. c_S has said that Mina is happy. | (locutionary act) |
| b. c_S has made a statement. | (illocutionary act) |
| c. c_S believes that Mina is happy. | (speaker's belief) |
| d. c_S wants c_A to believe that Mina is happy. | (speaker's preference) |
| e. $??c_A$ believes that Mina is happy. | (target situation) |

For this reason, it is odd for the speaker to deny the implicatures in (28a) through (28d),

¹⁰ Some of the implicatures discussed below may not be covered by the four major maxims mentioned in Grice (1975), but as he mentioned, the cooperative principle covers more than the four major maxims.

but not (28e), as shown in (29). Moreover, the addressee can argue against the truth of the statement, as in (30):

- (29) a. ??Mina is happy, but I have not said that Mina is happy.
 b. ??Mina is happy, but I have not made a statement.
 c. ??Mina is happy, but I do not believe Mina is happy.
 d. ??Mina is happy, but I do not want you to believe Mina is happy.
 e. Mina is happy, but you do not believe so.
- (30) A: John is happy.
 B: No, he is not.

The reason is that John's happiness is a matter of an established fact before the utterance, and the addressee might believe or know that John is not happy and argue against the statement. Only if the statement is not argued against by the addressee does it update the common ground.

In (28), (28a) and (28b) come from the utterance of the sentence, not from the sentence itself, and they cannot be considered semantic meanings. In (28c) and (28d), the speaker intends to share with the addressee the information that Mina is happy. If the addressee accepts it, the common ground is updated with the proposition that Mina is happy. (28e) is the result we can expect only after the addressee accepts the meaning of the sentence. Therefore, what the speaker conveys only through the sentence is that Mina is happy and it can be the semantic meaning of the sentence.

Under the context $c = \langle c_S, c_A, c_I, c_{CG} \rangle$, uttering an imperative results in a situation where the speaker has issued a command. A command involves an act on the part of the addressee, and it may not be accepted by the addressee. However, whether the addressee accepts it or not, the speaker makes two things clear, in addition to the basic locutionary and illocutionary acts. The speaker believes the content must be satisfied and wants the addressee to believe the same thing, as in (31c) and (31d):

- (31) By uttering " $!\varphi$ ", it becomes true
- | | |
|---|-------------------------|
| a. that c_S has made an utterance | (locutionary act); |
| b. that c_S has issued a command | (illocutionary act); |
| c. that c_S believes that φ | (c_S 's belief); |
| d. that c_S wants c_A to believe that φ | (c_S 's preference); |
| e. but not that c_A believes that φ . | (target situation) |

The implicature that the speaker has made an utterance reflects the locutionary act, and the implicature that the speaker has issued a command is an illocutionary act. The implicature (31b) indicates that an imperative is interpreted with the meaning of performativity. These two implicatures follow necessarily from the utterance itself if the speaker is sincere.¹¹ The content of the command is reflected in (31c) and (31d), where I assume that φ includes the meaning of modality. If the four implicatures in (31a) through (31d) are canceled, the discourse becomes odd. On the other hand, it is not implicated that the addressee (decides to) believe that φ . Therefore, the addressee may not accept the command.

This is illustrated in (32). For an imperative "Leave now!", we can confirm the implicatures predicted in (31) are correct:

- (32) a. ??I order you to leave now, but I have not made an utterance.
 b. ??I order you to leave now, but I have not issued a command.
 c. ??I order you to leave now, but I do not believe that you must leave.
 d. ??I order you to leave now, but I do not want you to believe that you must leave.
 e. A: Leave now!
 B: No, I won't.

To make the discourses more natural, I rephrased the imperative as an explicit performative. If the addressee accepts the imperative, the common ground is expected to be updated with the proposition that the addressee must leave. Note that the imperative is interpreted with the meaning of modality.

However, in order to discuss the semantics of imperatives, I need to discuss more about what properties imperatives have. There are many analyses of imperatives, but none of them showed that imperatives can change the actual world:¹²

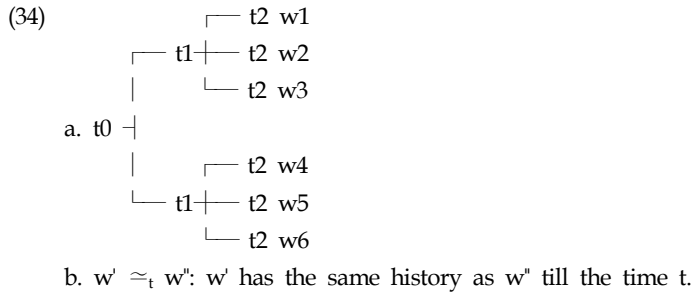
- (33) A: I know I must stay here until 5. But my brother is coming to see me at 4.
 B: Okay, you leave at 3!

11 There are cases where the speaker is insincere and makes a joke. In such cases, the speaker herself does not believe the truth of the corresponding modal statement.

12 There are three main previous analyses of imperatives: Portner (1997, 2004, 2005, 2007) claims that an imperative is an item in the To-do List for an agent. Schwager (2006) & Kaufmann (2012) claims that an imperative is simply a modal statement with some special presuppositions. Condoravdi and Lauer (2011, 2012) claim that an imperative contains a proposition that the speaker wants to be realized by the addressee, and that an imperative acts as an inducement for the addressee to make the proposition true. However, none of them tried to capture the property of imperatives to override presuppositions.

From A's statement, it is presupposed that A must stay at a particular place until 5. B's utterance, however, makes it true that A may leave at 3. From this discourse, we can make a couple of points. First, an imperative has the meaning of a modal statement as part of its meaning and the type of modality is determined contextually. In (33), the imperative is interpreted as a permission, which involves deontic possibility. This follows from the contextual fact that the previous obligation is revised. This can be compared with (32), where the imperative is normally understood as having the meaning of deontic necessity. Second, an imperative can override a presupposition. In the actual world, it was true before B's utterance that A must stay at a particular place until 5, but this presupposition is overridden by the imperative. Still the discourse is considered consistent. This means that an imperative can legitimately change the actual world by overriding some previous presuppositions.

To capture the property of imperatives that they can change the actual world, the meaning of an imperative needs to be defined so that the truth-value of the sentence can change. See Yeom (in progress) for more extensive discussions of this analysis of imperatives. The actual world changes over time. To deal with the semantics of an imperative formally, I follow Thomason's (1984) model of possible worlds that share common pasts and branch out into different possible futures:



The symbol " $\simeq_t$ " indicates a relation between two possible worlds that have the same history till t but have different futures. In the diagram, $w1 \simeq_{t1} w3$ holds, but $w1 \simeq_{t0} w5$ does not.

The content of a command should not be considered a proposition. Since a command is a world changer, it needs to be defined as a relation between possible worlds. In (33), it was true that A must stay until 5 in the actual world, but after B's utterance, it has become false. Instead, A can leave at 3. That is, the actual world changes. If an imperative " φ -la" (or, " $!\varphi$ " for convenience's sake) is uttered in a possible world w , the actual world becomes a possible world w'' such that φ is true in w'' . w'' is minimally different from w by the condition that

more propositions that are supported in the common ground are true in w'' than in any other φ -world w' that shares the same history with w till the utterance time. We can define the semantics of an imperative in two ways:

- (35) a. static version: $\llbracket !\varphi \rrbracket^{w,c} = \lambda w''[\mathbf{w}'' \in \max_{\text{cCG}}(\{w' \in W: w' \simeq_{\text{cT}} w \wedge \varphi'(w')\})]$
 $\max_{\text{cCG}}(X) = \{w' \in X: \text{for any other } w'' \in X,$
 $\{q: \text{cCG supports } q \wedge q(w'')\} \subseteq \{q: \text{cCG supports } q \wedge q(w')\}\}$
 b. dynamic version: $c + !\varphi = \cup \{\max_{\text{cCG}}(\{w' \in W: w' \simeq_{\text{cT}} w \wedge \varphi'(w')\}): w \in \text{cCG}\}$

Since we need to consider a context in interpreting an imperative and we need to deal with all types of sentences together, we prefer the dynamic version, in which all types of sentences can be interpreted as contextual change potentials.¹³ The actual world changes to a possible world in which φ is true but there may be more than one possible world in which φ is true. Among those possible worlds, the actual world is expected to change minimally. For this reason, the new possible world(s) need(s) to be closest to the old possible world, and the criteria are the information in the common ground.¹⁴ A possible world is selected which is most compatible with the information in the common ground.

On the other hand, if an imperative is embedded as a *ko*-phrase selected by the verb *malha* 'say', the implicatures (31c) and (31d) can be canceled:

- (36) mina-nun inho-eykey ttena-la-ko malha-yess-una sasil inho-ka
 Mina-top Inho-to leave-imp-cmp say-pst-but in.fact Inho-nom
 ttena-yaha-n-ta-ko sayngkakha-ci anh-ass-ta.
 leave-must-npst-dec-cmp think-nml not.do-pst-dec
 'Mina told Inho to leave, but she did not think that he must leave.'
 (37) mina-nun inho-eykey ttena-la-ko malha-yess-una sasil inho-ka
 Mina-top Inho-to leave-imp-cmp say-pst-but in.fact Inho-nom
 ttena-nun kes-ul wenha-ci anh-nun-ta.
 leave-adn thing-acc want-nml not.do-npst-dec
 'Mina told Inho to leave, but she does not want him to leave.'

13 In a static semantics, a declarative sentence is interpreted as a proposition, but an imperative is as a relation between possible worlds and an interrogative is as a function of partitioning possible worlds.

14 If a sentence ϕ is compatible with a common ground, ϕ is true or false in an epistemic alternative in the common ground. If it is true in w , the set of ϕ -worlds that are maximally close to w is the set of epistemic alternatives in which ϕ is true. Therefore, in the interpretation in (35a), w and w' are in the common ground and ϕ is true in both. Even if ϕ is not true in w , the set of ϕ -worlds maximally close to w is also the set of epistemic alternatives in which ϕ is true. Therefore, if ϕ is compatible with the common ground, it has the effect of updating the common ground.

This indicates that if a *ko*-phrase is used, the speaker is not committed to the content of the imperative. For this reason, the content of the *ko*-phrase can be considered a joke:

- (38) mina-nun inho-eykey ttena-la-ko malha-yess-una sasil kukes-un
Mina-top Inho-to leave-imp-cmp say-pst-but in.fact it-top
nongtam-i-ess-ta.
joke-be-pst-dec
'Mina told Inho to leave, but it was a joke.'

This follows if a *ko*-phrase is not interpreted as denoting an extensional semantic entity.

This is related to the fact that if a *ko*-phrase is selected by a conveying verb, the verb is not used as a performative verb. Consider the following examples:

- (39) a. A: inho-nun ettehkey myenglyengha-yess-nya?
Inho-top how order-pst-int
'How did Inho order?'
B: ku-nun motwu kitali-la-ko myenglyengha-yess-ta.
he-top everyone wait-imp-cmp order-pst-dec
'He ordered (us) that everyone should wait.'
b. inho-nun mina-lul manna-la-ko ha-ci anh-ko caki tongsayng-ul
Inho-top Mina-acc meet-imp-cmp say-nml not.do-and his sister-acc
manna-la-ko ha-yess-ta.
meet-imp-cmp say-pst-dec
'Inho did not tell me to meet Mina, but to meet his sister.'
- (40) inho-nun mina-eykey 3 si-ey ttena-l kes-ul 4 si-ey
Inho-top Mina-to 3 o'clock-at leave-adnm thing-acc 4 o'clock-at
ttena-la-ko malha-yess-ta.
leave-imp-cmp say-pst-dec
'Inho told Mina of having to leave at 3 as though she should leave at 4.'

In (39a), the *ko*-phrase is mentioned as an answer to the question of *ettehkey* 'how'. This indicates that the *ko*-phrase is not the syntactic object of the verb *myenglyengha* 'order'.¹⁵ The question word is used when the reference of the *ko*-phrase does not matter. What matters is

15 One anonymous reviewer mentions the possibility that *ettehkey* 'how' can be replaced with expressions like *mwue-la-ko* 'what-be-cmp' or *mwues-ul* 'what-acc'. *Mwue-la-ko* is not helpful because it is also a *ko*-phrase. To me, *mwues-ul* does not seem acceptable.

the expression used in the *ko*-phrase. This leads to the observation in (39b). In (39b), *mina* in the first embedded clause is replaced with *caki tongsayng* ‘his sister’ and the two embedded imperatives are considered different even if Mina is Inho’s sister. This indicates that the two expressions are embedded in two different intensional semantic entities, which are introduced by the use of *ko*. In (40), the *kes*-phrase is understood as an actual thing to do and the *ko*-phrase is understood as a command that is not correct. The command that is not correct cannot be interpreted with respect to the actual world. It can be captured by some non-actual possible worlds that are compatible with the content of the incorrect command. The non-actual possible worlds are introduced by an intensional semantic object. Therefore, the *ko*-phrase is interpreted as an intensional semantic object. Similar observations are made in embedded declaratives in Yeom (2018).

If a sentence with an embedded imperative is to be understood as a command performative, the embedded clause needs to be interpreted with respect to the actual world so that it can be understood as if it were an imperative as a root clause, which is interpreted with respect to the actual world. As we saw, however, a *ko*-phrase cannot be interpreted with respect to the actual world. This is the reason that a sentence with a *ko*-phrase cannot be used in a command performative, as we saw (9) above.

Since a *ko*-phrase is never used as part of a command performative, we can assume that a *ko*-phrase is interpreted as an intensional semantic entity by abstracting over possible worlds in (35a):

$$(41) \llbracket \varphi\text{-la-ko} \rrbracket^w = \llbracket \wedge \varphi\text{-la} \rrbracket^w = \lambda w' \llbracket \varphi\text{-la} \rrbracket^{w'} = \lambda w' \lambda w'' [\langle w', w'' \rangle \in \llbracket \varphi\text{-la} \rrbracket], \text{ where } \varphi \text{ is an imperative.}$$

Here $\llbracket \varphi\text{-la} \rrbracket^w$ is the extensional meaning given in (35a). The meaning of a *ko*-phrase with an imperative is an abstraction over the possible world w . And verbs that take a *ko*-phrase do not generate any implicatures that entail the speaker is committed to the content of the *ko*-phrase. They just convey the meaning that the agent utters an expression that has the (intensional) meaning of an imperative:

$$(42) \llbracket \alpha\text{-la-ko malha} \rrbracket^w = \lambda r \lambda x \exists s \exists e [\text{expression}(w)(s) \ \& \ \text{say}(w)(s)(x)(e) \ \& \llbracket \alpha\text{-la} \rrbracket = r]$$

Here r is a relation between possible worlds given in (41).¹⁶ I assume that the meaning of *malha* ‘say’ reflects the existence of an event of uttering a sentence, and that there is no further

16 If α is a declarative, $\llbracket \alpha \rrbracket$ is a proposition.

semantic implication on the speaker's or addressee's belief or preference, as shown in (36) and (37).

3. *Kes*-phrases and command performatives

I showed that a *ko*-phrase cannot be used in a command performative. To show how a *kes*-phrase can be used in a command performative, I need to discuss various things. First, what is the meaning of an explicit performative? Then I need to discuss what a *kes*-phrase means, in contrast with a *ko*-phrase.

3.1. Meanings of explicit performative sentences

To get the idea of what an explicit performative sentence means, it is helpful to look at its pragmatic meanings. This is especially necessary because a performative sentence relies on the utterance context. The explicit performative in (43) causes all the implicatures in (44):

(43) chwukhaha-n-ta.

congratulate-npst-dec

'I congratulate you.'

(44) a. c_S has said that (s)he has congratulated.

b. c_S has congratulated.

c. c_S believes that (s)he has congratulated.

d. c_S wants c_A to believe that c_S has congratulated.

e. c_A believes that c_S has congratulated.

One crucial difference from a declarative and an imperative is (44e). Here (44e) is an additional fact that arises because the utterance is an explicit performative. Before this sentence is uttered, it is not an established fact whether the speaker will congratulate, but the utterance adds a fact to the actual world. It is odd for the addressee to refute the fact, as shown in (45). It becomes a fact at the utterance context at the presence of the speaker and the addressee because the act of uttering is also an act of congratulating. Both the speaker and the addressee know this. That is, it automatically becomes part of the common ground. This is what is observed by Lemmon (1962), Hedenius (1963), Bach & Harnish (1979), Ginet (1979), Bierwisch (1980), Leech (1983), Jary (2007), among others.

- (45) A: chwukhaha-n-ta.
 congratulate-npst-dec
 'I congratulate you.'
- B: ??ani, ne-nun chwukhaha-ci anh-a.
 no you-top congratulate-nml not.do- dec
 'No, you do not.'

Even if an explicit performative is embedded, the extra illocutionary act is still implicated.¹⁷ In (46), the act of congratulating is still implicated.

- (46) ??inho-nun chwukhaha-n-ta-ko malha-yess-ta. kulena ku-nun
 Inho-top congratulate-npst-dec-cmp say-pst-dec but he-top
 chwukhaha-ci anh-ass-ta.
 congratulate-nml not.do-pst-dec
 'Inho said he congratulated but he did not congratulate.'

This shows that the core property of explicit performatives is maintained even in embedded clauses.

This phenomenon comes from the definitional property of explicit performatives that if a sentence is uttered, the speaker has done the act denoted by the verb of the matrix clause:

¹⁷ Some implicatures we get from performatives as root clauses are not maintained if they are embedded. If the same sentence is indirectly conveyed, the speaker of the embedded clause does not have any commitment related to the content of the embedded clause. Thus the third and fourth implicatures in (28) are not ensured:

- i. inho-nun mina-ka hayngpokha-ta-ko malha-yess-ta.
 Inho-top Mina-nom happy-dec-cmp say-pst-dec
 'Inho said that Mina was happy.'
- ii. kulena ku-nun mina-ka hayngpokha-ta-ko sayngkakha-ci anh-nun-ta.
 but he-top Mina-nom happy-dec-cmp think-nml not.do-npst-dec
 'But he did not think that Mina was happy.'
- iii. kulena ku-nun nay-ka mina-ka hayngpokha-ta-ko sayngkakha-ki-lul pala-ci
 but he-top I-nom Mina-nom happy-dec-cmp think-nml-acc want-nml
 anh-nun-ta.
 not.do-npst-dec
 'But he did not want me to think that Mina was happy.'

Considering these observations, we need thorough examinations of the differences of implicatures between root clauses and embedded clauses.

- (47) A sentence is an explicit performative iff if it is uttered, the speaker has done the act expressed by the matrix verb of the sentence.

This can be considered a core property of a performative verb.

Another property of explicit performatives is that they can change the actual world. Performative sentences are not about established facts before the utterance. They express acts that speakers do when they are uttered and the acts are carried out by the speaker's decision. Since the speaker makes a decision at the utterance time, the utterance can be contradictory with a previously established fact:

- (48) A: You never say thank you.

B: Thank you for telling me that. Nobody's ever told me that. I will change.

In the discourse, A characterizes B as a person who never says thank you. B realizes and admits this now but B decides to change and say thank you. However, the discourse is not considered contradictory. An explicit performative can legitimately convey something inconsistent with a previously established fact. In the example above, the proposition that B never says thank you was true, but now it becomes false by B's utterance. Therefore, the actual world has changed over time.¹⁸

Since an explicit performative can change the actual world, we can interpret an explicit performative just like an imperative:¹⁹

- (49) a. static version: $\llbracket \varphi_{[\text{performative}]} \rrbracket^c$
 $= \lambda w \lambda w' [w' \in \max_{\text{CG}}(\{w' \in W: w' \simeq_{\text{CT}} w \wedge \varphi'(w')\})]$
 b. dynamic version: $\text{c}_{\text{CG}} + \varphi_{[\text{performative}]}$
 $= \cup \{\max_{\text{CG}}(\{w' \in W: w' \simeq_{\text{CT}} w \wedge \varphi'(w')\}): w \in \text{c}_{\text{CG}}\}$

18 In contrast with performative sentences, descriptive sentences are about established facts. Established facts are reflected in the common ground, and all established facts need to be consistent. For this reason, each new descriptive statement must keep the common ground consistent.

19 Krifka (2014) proposes the semantics of an explicit performative as the type of css , where c is the type of a context and s is the type of a world-time index, but his proposal is not yet based on sufficient empirical evidence. And when he says that an explicit performative changes one index to another minimally, I am not sure what he means by the minimal change from one index i to another i' . He seems to assume that all explicit performatives minimally change possible worlds, but I do not agree with him on this. Adding a new fact by an explicit performative does not necessarily mean the actual world changes. If the new fact is compatible with the common ground, we do not need to assume that the actual world has changed. If the actual world has changes, we need a context revision. But he just assumes a context update in most cases.

This means that a performative changes a possible world w to another possible world w'' in which the content of the performative sentence is a (possibly new) fact and which is maximally close to w with respect to the propositions supported in the common ground. See (35a) for this condition. If the performative is compatible with the common ground, it is expected to update the common ground with the new fact. But if it is incompatible with the common ground, it will introduce a new common ground by changing each possible world in it to (a) different possible world(s).

3.2. *Kes*-phrases and command performatives

Command performatives are different from imperatives in two respects. One is that imperatives are used with various meanings like command, permission, warning, plea, wish, advice, invitation, concession, etc., as pointed out in Schmerling (1982), Davies (1986), Palmer (1986), Bybee et al. (1994), and Xrakovskij (2001), Schwager (2006), Kaufmann (2012), etc. We saw cases where an imperative is understood as a command and permission. Here are more examples:

- (50) a. A: Excuse me! How can I get to the station? (Advice)
 B: Take a number 10 bus!
 b. Be happy! (Wish)
 c. Okay, then sue me, if you have to! (Concession)

This means that an imperative has the meaning of modality and the modality is determined contextually. On the other hand, the performative use of the verb *myenglyengha* 'order' invariably has the meaning of command. In this respect, the performative verb determines the type of modality. The other is that imperatives are inherently performative and they cannot be used descriptively. On the other hand, a verb which can be used in an explicit performative may not be used in a performative sentence. Only if the verb is used as a performative verb does it have the effect of a command performative. Therefore, performativity does not come from the verb itself.

Despite these differences, we can assume that an imperative and a command performative with *myenglyengha* 'order' have the same meaning. I will show how an explicit performative can have the same meaning as an imperative. We saw in (6) that a verb like *myenglyengha* 'order' can take a *kes*-phrase and that if it is used as a performative verb, the whole sentence becomes a command performative with the content of the *kes*-phrase. This means that the semantics of a *kes*-phrase is different from that of a *ko*-phrase.

When a *kes*-phrase can be the content of a command performative, it has an adnominal clause ending with *-(u)l*:

- (51) na-nun ne-eykey tangcang ttena-l kes-ul myenglyengha-n-ta.
I-top you-to at.once leave-adnm thing-acc order-npst-dec
'I order you to leave at once.'
- (52) na-nun ne-ka tangcang ttena-l kes-ul myenglyengha-n-ta.
I-top you-to at.once leave-adnm thing-acc order-npst-dec
'I order you to leave at once.'

To understand why *-(u)l* is used, we need to consider various *kes*-phrases and other phrases as complements of various performative verbs.

It seems that the form of an adnominal ending is selected by the verb that takes the *kes*-phrase. A verb like *myenglyengha* 'order' or *yokwuha* 'demand' selects for the ending of *-(u)l*, while a verb like *helakha* 'allow' or *kumciha* 'forbid' selects for the ending of *-nun*.²⁰

20 *kes* can be preceded by three forms of adnominal clauses: *-(u)l*, *-nun*, *-(u)n*. *-nun* is attached to a non-stative verb and expresses the aspect of imperfective. *-(u)n* is attached to a non-stative verb to express perfectiveness or attached to a stative verb. Neither has the meaning of modality. *-(u)l* is attached to any verb to express modality:

- | | |
|--|---|
| i. a. mek-nun kes
eat-adn thing
'eating' | b. sa-nun kes
buy-adn thing
'buying' |
| ii. a. kem-un kes
black-adn thing
'being black' | b. coyongha-n kes
quiet-adn thing
'being quiet' |
| iii. a. mek-un kes
eat-adnp thing
'having eaten' | b. sa-n kes
buy-adnp thing
'having bought' |
| iv. a. kem-ess-te-n kes
black-pst-rtr-adnp thing
'having been black' | b. coyongha-yess-te-n kes
bitter-pst-rtr-adnp thing
'having been quiet' |
| v. a. mek-ul kes
eat-adnm thing
'to eat' | b. sa-l kes
buy-adnm thing
'to buy' |
| c. kem-ul kes
black-adnm thing
'to be black' | d. coyongha-l kes
quiet-adnm thing
'to be quiet' |

In this paper I am concerned with imperatives or command performatives. Thus, we are only concerned with the selection of *-(u)l* and *-nun*.

- (53) a. na-nun ney-ka ttena-{l, *?nun} kes-ul myenglyengha-n-ta.
 I-top you-nom leave-{adnm, adni} thing-acc order-npst-dec
 'I order you to leave.'
- b. na-nun ney-ka ttena-{l, *?nun} kes-ul yokwuha-n-ta.
 I-top you-nom leave-{adnm, adni} thing-acc demand-npst-dec
 'I demand you to leave.'
- (54) a. na-nun ney-ka ttena-{??l, nun} kes-ul helakha-n-ta.
 I-top you-nom leave-{adnm, adni} thing-acc order-npst-dec
 'I allow you to leave.'
- b. na-nun ney-ka ttena-{??l, nun} kes-ul kumciha-n-ta.
 I-top you-nom leave-{adnm, adni} thing-acc forbid-npst-dec
 'I forbid you to leave.'

Whether to use *-(u)l* or *-nun* is an issue we need to pursue in a separate paper, but we can give the following tentative generalization, which is sufficient for the purpose of this paper:²¹

- (55) α -(*u*)l kes is used if it denotes a proposition where α describes something that will be done in the future relative to the reference time of the embedding clause.

This does not mean that *-(u)l* itself determines the type of modality. The verbs *myenglyengha* 'order' and *yokwuha* 'demand' require deontic modality, but the adnominal ending can simply express the meaning of futurity without deontic modality:

21 There are cases where *-(u)l* has no meaning of modality:

- i. mina-nun wus-{ul, *(n)un} {ttalum, ppwun}-i-ess-ta.
 Mina-top laugh-{adnm, adn} {only, only}-be-pst-dec
 'Mina only laughed.'
- ii. mina-nun inho-ka wus-{ul, *(n)un} ttay o-ass-ta.
 Mina-top Inho-nom laugh-{adnm, adn} time come-pst-dec
 'Mina came when Inho was laughing.'

However, this is a case where no other adnominal ending is allowed. If other adnominal endings are also allowed, *-(u)l* is used when it had the meaning of modality:

- iii. inho-ka cip-ulo tolaka-{l, n, nun} seym-i-ess-ta.
 Inho-nom home-to return-{adnm, adnp, adni} reckoning-be-pst-dec
 'Inho {was going to return, as good as returned, is as good as returning} home.'

The use of *-(u)l* gives the meaning of future-oriented modality.

- (56) inho-nun mina-ka o-l kes-ul {mit, al}-e/ass-ta.
 Inho-top Mina-nom come-adnm thing-acc {believe, know}-pst-dec
 'Inho {believed, knew} that Mina would come.'

To cover these cases, I will assume that $-(u)l$ is an adnominal ending that has the meaning of modality as well as tense and that the type of modality is determined in the linguistic or pragmatic context. The verb that selects for an adnominal clause ending with $-(u)l$ constitutes a linguistic context.

Considering the semantics of $-(u)l$, we can interpret a *kes*-phrase as follows:

- (57) $\llbracket [_{VP} \alpha] \rrbracket = \lambda w \lambda e \llbracket \alpha \rrbracket (w)(e)$
 $\llbracket [_{Asp} \varnothing] \rrbracket =$ (i) $\lambda P \lambda w \lambda t [\tau(e) \sqsubseteq t \wedge P(w)(e)]$ if the VP is eventive;
 (ii) $\lambda P \lambda w \lambda t [t \sqsubseteq \tau(e) \wedge P(w)(e)]$ if the VP is stative.
 $\llbracket [_{TP} \alpha-(u)l] \rrbracket^c = \lambda w [\forall w' [w' \in \mu(w, c_T) \rightarrow \exists s \exists t' [c_T < t' \wedge \alpha'(w')(t')]]]$, where
 μ is a modal operator determined contextually.
 (58) $\llbracket kes \rrbracket = \lambda P [P]$

A VP denotes a property of events and an Asp(ect) converts it into a property of times. I assume that $-(u)l$ is interpreted with the meaning of modality with respect to the local context c , which also provides a reference time. The adnominal clause combines with *kes*, which is semantically null and is a defective noun in that it requires an adnominal clause. Therefore, *kes* heads an NP but the NP has the same meaning as the adnominal clause.²²

To get a clue to the meaning of the verb *myenglyengha* 'order', I will consider a case where the verb takes an NP argument:

- (59) cangkwn-un chelswu-lul myenglyengha-yess-ta.
 general-top retreat-acc order-pst-dec
 'The general ordered a retreat.'

In the example, the verb takes an NP headed by a common noun *chelswu* 'retreat'. This sentence is supposed to have the following meaning:

- (60) $\text{ordered}(w)(\text{the_general}, c_A, \forall w' [w' \in \text{DEON}(w, t) \rightarrow \text{retreat}(w')(c_A)])$

²² One additional meaning is that a *kes*-phrase denotes a non-person semantic entity, but I will ignore it. See Yeom (2015) for extensive discussions on adnominal clauses and *kes*-phrases.

The NP argument of the verb is interpreted in the scope of the modal operator that is introduced by the use of the verb. This indicates that the verb *myenglyengha* ‘order’ inherently selects for an NP with the meaning of modality, and if the NP does not have the meaning of modality, the verb provides the type of modality for the interpretation of the NP.

A difference between a noun like *chelswu* ‘retreat’ and a *kes*-phrase is that the latter includes an adnominal clause with a modal element. If *myenglyengha* ‘order’ selects a *kes*-phrase, the requirement of modality is satisfied. However, the adnominal ending *-(u)l* does not determine the type of modality. Instead, the verb requires the *kes*-phrase to be interpreted with deontic necessity modality.

Even if this is the case, there is still one issue to resolve. If the verb *myenglyengha* is used as a performative verb, the *kes*-phrase is supposed to be interpreted with respect to the actual world. In (40), we saw that the *kes*-phrase expresses the content of a real command and the *ko*-phrase represents the content of a non-real content. This can be confirmed by the following example:

- (61) ??in_{ho}-nun mina-lul manna-l kes-ul myenglyengha-ci anh-ko caki
 In_{ho}-top Mina-acc meet-adnm thing-acc order-nml not.do-and his
 tongsayng-ul manna-l kes-ul myenglyengha-yess-ta.
 sister-acc meet-adnm thing-acc order-pst-dec
 ‘In_{ho} did not tell me to meet Mina, but to meet his sister.’

Here the main verb is not in performative use. If Mina is In_{ho}’s sister, the sentence sounds odd. This indicates that the two *kes*-phrases are interpreted with respect to the actual world and that they denote the same thing. This makes the sentence contradictory. If the two *kes*-phrases were interpreted intensionally, the sentence would be fine, as in (39b). The fact that a *kes*-phrase refers to something that needs to be done in the actual world makes it possible for a *kes*-phrase to provide the content of the command if the embedding clause is used as a performative, as in (40).

Now we are ready to interpret a sentence which is used as a performative sentence, as in (6). The sentence is repeated here for convenience’s sake. The basic meaning of *myenglyengha* is given in (63). If the verb is in performative use and the sentence is a command performative, the sentence has the meaning given in (64):

- (62) na-nun ne-eykey tangcang ttena-l kes-ul myenglyengha-n-ta.
 I-top you-to at.once leave-adn thing-acc command-npst-dec
 ‘I order you to leave at once.’

$$(63) \lambda w \lambda p \lambda x \lambda y [\text{order}'(w)(y, x, \forall w' [w' \in \text{DEON}(w) \rightarrow p(w')])]$$

$$(64) c + [\alpha\text{-ul myenglyengha}_{[\text{performative}]}]$$

$$= \bigcup \{\max_{\text{CCG}}(\{w': w' \simeq_{\text{CT}} w \wedge \text{order}(w')(c_S, c_A, \alpha'(w'))\}): w \in \text{CCG}\}$$

The verb *myenglyengha* ‘order’ takes the extensional meaning of the *kes*-phrase, which is given as $\alpha'(w)$ in (64). In this interpretation, w' is a variable for epistemic alternatives, each of which is a candidate for the actual world.

The question is what the meaning of “order” in (64) is when the sentence is performative. As I previously mentioned, the performative verb determines the modality in the meaning of the *kes*-phrase taken by the verb. The verb *order* makes the *kes*-phrase interpreted with deontic modality. The utterance of a command performative with the verb *order* generates the following implicatures:

- (65) a. c_S has uttered an explicit performative.
 b. c_S has issued an order.
 c. c_S believes that c_A must do something so that the content of the *kes*-phrase is realized.
 d. c_S wants c_A to do the same thing.

Here (65a) and (65b) are not refutable. (65a) is simply a fact of utterance. (65b) comes from the use of the performative verb. This has two effects. One is that it is used as a performative verb and interpreted as a world changer. I suppose that this has the same effect as the mood marker of the corresponding imperative. This meaning is already reflected in (64). The other effect is that the verb determines the type of modality in the meaning of the *kes*-phrase. That is, the *kes*-phrase is interpreted with the meaning of deontic necessity. And if the addressee accepts the content of the *kes*-phrase, the explicit performative becomes a valid command. Therefore (65c) and (65d) reflect the content of the command, and we can assume that the formula with “order” in (66) has the following meaning:²³

$$(66) \text{order}(w')(c_S, c_A, \alpha'(w')) = \alpha'(w'), \text{ where } \alpha' \text{ includes the modality of deontic necessity.}$$

This means that an explicit performative with “order” has the effect of imposing the modality of deontic necessity in the meaning of the *kes*-phrase. When the verb *myenglyengha*

23 If the complement is an event denoting NP, it needs to be interpreted with the meaning of modality:

i. $\text{order}(w')(c_S, c_A, \alpha'(w')) = \forall w' [w' \in \text{DEON}(w, t) \rightarrow \text{retreat}(w')(c_A)]$

'order' takes a *kes*-phrase, the performative in (62) can be interpreted as follows:

$$(67) \text{ c + (62) = } \\ \cup \{ \max_{\text{cCG}}(\{w': w' \simeq_{\text{cT}} w \wedge \forall w''[w'' \in \text{DEON}(w', \text{cT}) \rightarrow \text{leave}(w'')(c_A)]\}) : w \in \text{cCG} \}$$

This is how a performative with a *kes*-phrase can be interpreted as an imperative.

4. Conclusion

In this paper, I showed that if a verb takes a *ko*-phrase, the *ko*-phrase is always interpreted as denoting an intensional semantic entity. For this reason, if a conveying verb takes a *ko*-phrase, the sentence cannot be used as a command performative. On the other hand, if a conveying verb can take a *kes*-phrase, the sentence can be used as a command performative. In this case, the *kes*-phrase is interpreted into an extensional semantic entity. This makes it possible for the *kes*-phrase to be understood as the content of the command.

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